Processing Quantifier Scope Interpretations in Mandarin Chinese

The interpretation of scope ambiguity in doubly quantified sentences has been important in linguistic theory in English and Mandarin (S. F. Huang, 1981; C.-T. J. Huang, 1982; Kurtzman & MacDonald, 1993). The *Every…one…* structure (e.g., *每个男孩爬了一棵树* 'Every boy climbed a tree' (hereafter **EVERY**) may have either the surface scope interpretation (i.e., the 'universal-wide' reading) that "each boy climbed a different tree", or the inverse scope reading (i.e., 'existential-wide' reading) that "all boys climbed the same tree." Syntactically, quantifier raising (May, 1977) has been proposed to derive these two interpretations through moving the existential objects to a position higher than the universal subject. Semantically, entailment accounts (Aoun & Li, 1989; Bruening, 2001; Hornstein, 1995) consider inverse reading entails the surface one. Changing the quantifiers of subject and object, i.e., *One…every…* structure, as in *一个男孩爬了每棵树* 'One boy climbed every tree' (hereafter **ONE**) may lead to similar ambiguities, though much little research has focused on it.

Unlike English, Mandarin has been argued to be scope-rigid theoretically (Huang 1981; Lee 1986; Aoun & Li 1989; 2003), allowing only the surface scope. This aligns with predictions from accounts of Semantic Entailment (Aoun & Li, 1989; Bruening, 2001; Hornstein, 1995), Processing Scope Economy (Anderson, 2004), and the Isomorphic Principle (Aoun & Li, 1989), all favoring the surface scope readings. In addition to quantifiers' types and their relative orders, prior research also suggests that verb types (Kurtzman & MacDonald, 1993, Zhou & Gao, 2009) together with aspect markers may influence scope processing (Hsu & Lin, 2014; 2016). Yet, prior studies mostly examined *EVERY*-type and not *ONE*-type sentences.

To address this research gap, two self-paced reading experiments were conducted to (1) examine the effects of verb types (action and perception) and aspect types on interpreting *EVERY*-sentences while controlling for contextual bias, and (2) test the scope preferences for *EVERY* and *ONE* sentences using action and perception verbs in their bare form (Tables 1 & 2).

Experiment 1 (n=80, mean age=23.10; female = 45) asked participants to read an EVERY target and to judge whether the follow-up sentence expresses the target's meaning on a 7-point Likert scale (1=totally unrelated; 7=totally related), through 192 sentence-pairs (24 verbs x 4 aspects x 2 determiners). The ratings of semantic relatedness showed that *le* was the highest (82.92%) and *guo* was the lowest (76.88%), consistent with Collart and Chan's study (2021) that *le* and *guo* encode past reference via different mechanisms. We then fitted linear mixed effect models to the log-transformed reaction time (RT) of whole trial (Reg1-13). Results showed that action verbs were read significantly slower that perception verbs (β=-0.03, se=0.01, p=0.01). Aspect markers significantly increased whole-RTs relative to bare verbs (*le*: β=0.05, se=0.02, p=0.003; *zhe*: β=0.05, se=0.02, p=0.003; *guo*: β=0.07, se=0.02, p<0.001), although differences among the three aspects were not significant (ps>0.14). In the verb and aspect positions (Reg 3+4; Fig. 1), verb with aspect elicited significantly longer RTs (*le*: β=0.63, se=0.05, p<0.001; *zhe*: β=0.64, se=0.05, p<0.001, *guo*: β=0.72, se=0.05, p<0.001), whereas verb type showed no significant main effect or interaction with aspects (ps>0.12).

Experiment 2 (n=40, mean age=24.20; female = 27) asked participants to explicitly select singular or plural subject in the follow-up sentence, revealing their scope interpretation, with total 48 sentence pairs (24 verbs x 2 quantifier orders). 75% of items were preferred with a surface scope, while 25% preferred an inverse scope. ONE elicited a robust surface-scope preference (β=-10.26, se=1.24, p<0.001), but EVERY had no significant differences on either scope (surface=50.21%, inverse=49.79%). No significant effect was found for verb type (β=0.26, se=2.38, p=0.91). For RTs (Fig. 2), none of the factors elicited significant main effects (ps>0.15). However, in Reg Question1, inverse scope was significantly faster compared to surface scope under order EVERY (β=-0.20, se=0.07, p=0.005). **These findings suggest that** sentences with EVERY subject equally allow either scope interpretations, and that the interpretive difficulty is influenced by quantifier order, together with verb type and aspect.

Table 1. An example of Exp.1 sentences' regions

Doubly quantified sentence						Continuation						
Reg1 (Q1)	Reg2 (N1)	Reg3 (V)	Reg4 (Asp)	Reg5 (Q2)	Reg6 (N2)	Reg7 **(Det)**	Reg8 (N)	Reg9	Reg10	Reg11	Reg12	Reg13
每个	学生	清洗	了	一个	瓶子。	**这些**	瓶子	本来	瓶身	都	是	污垢。
every	student	wash	perf	one	bottle.	**these**	bottles	originally	body-of-bottle	fully	be	dirt.
Every student washed one bottle.						The bodies of these bottles were full of dirt.						

Note. Reg3 includes action and perception verbs; Reg4 includes bare form (skip it in exp.), aspects *le, zhe, and guo*; Reg7 includes singular and plural determiners.

Table 2. An example of Exp.2 sentences' regions

Doubly quantified sentence					Continuation					
Reg1 (Q1)	Reg2 (N1)	Reg3 (V)	Reg4 (Q2)	Reg5 (N2)	Question1 (**Det +** N)	Reg8 (N)	Reg9	Reg10	Reg11	Reg12
一个	学生	清洗	每个	瓶子。	**a. 这个**学生 b. **这些**学生	受到	了	老师	的	表扬。
one	student	wash	every	bottle.	**This / These** students	receive	perf	teacher	-'s	praise.
One student washed every bottle.					This student/ these students received the teacher's praise.					

Note. Reg1 & 4 are paired with either *one…every…* or *every…one…*; Reg3 includes action and perception verbs.

Figure 1. logRTs for Reg3+4, by Verb type and Aspect.

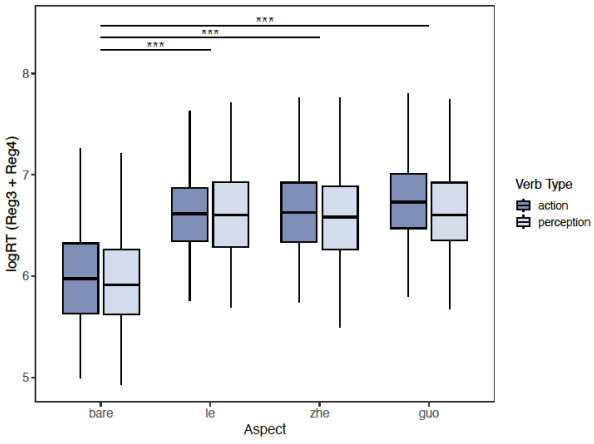
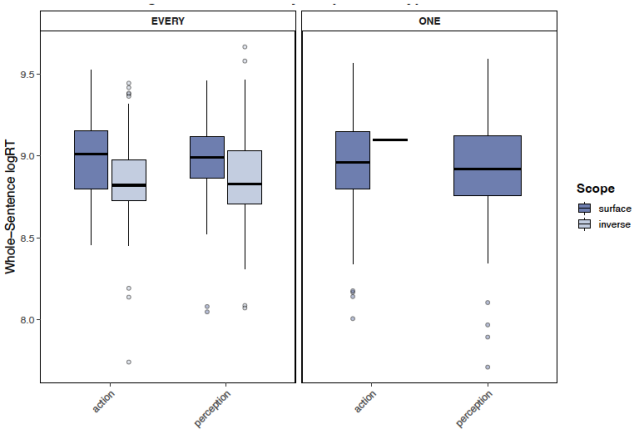


Figure 2. Whole-sentence logRTs by Quantifier order, Scope and Verb type.



Reference

- Anderson, C. (2004). The structure and real-time comprehension of quantifier scope ambiguity. Evanston, IL: Northwestern University dissertation.
- Aoun, J., & Li, Y. H. A. (1989). Scope and constituency. Linguistic inquiry, 20(2), 141-172.
- Bruening, B. (2001). QR Obeys Superiority: Frozen Scope and ACD. Linguistic Inquiry.
- Collart, A., & Chan, S.-h. (2021). Processing past time reference in a tenseless language: An ERP study on the Mandarin aspectual morphemes -le and -guo. Journal of Neurolinguistics, 59, Article 100998. https://doi.org/10.1016/j.jneuroling.2021.100998
- Huang, S. F. (1981). On the scope phenomenon of Chinese quantifiers. Journal of Chinese Linguistics, 9, 226–243.
- Hsu, Y.-Y., & Lin, C.-J. Charles. (2014). Verb Types, Contexts and Quantifier Scope Interpretation in Mandarin. The 22nd Annual International Association of Chinese Linguistics (IACL 22) and the 26th North American Conference on Chinese Linguistics (NACCL-26). University of Maryland, College Park, May 2-4, 2014.
- Zhou, P., & Gao, L. (2009). Scope Processing in Chinese. J Psycholinguist Res 38, 11–24. https://doi.org/10.1007/s10936-008-9079-x